

Final Year Project Report

Konvert



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Dedication

Final Approval

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Acknowledgment

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Our sincerest gratefulness goes to our advisor, Mr. Hafiz Ahsan Arshad, for his specialized mentorship and unwavering support. His guidance was necessary in helping us navigate the complications of enterprise-position software development and AI integration.

A special and significant note of appreciation is extended to the CEO of Bristol. We are truly recognized by the direct support and trust he placed in us to break a critical, real-world business problem for his association. His amenability to give us direct access to his company's functional perceptivity was the foundation of this design. We also admit the team at Almashreqs for their cooperation in easing the integration between the mobile ecosystem and the being web structure, icing data thickness across the platform.

Eventually, we fete the cooperative spirit of every group member — Asim, Obaid, and Farheen. This design is a testament to our collaborative fidelity to creating a localized result that not only fulfills our degree conditions but also serves as a point of pride for our country's technological geography.

Project Title:	Konvert
Objective	To design and develop an AI-integrated business management and sales tracking ecosystem that optimizes multi-tier enterprise hierarchies, streamlines distribution workflows, and provides a localized, secure alternative for data-driven decision making.
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Supervised by	Hafiz Ahsan Arshad
Starting Date	
Completion Date	
Tools Used	Flutter, Firebase, Android Studio, Google Maps API, TensorFlow and Scikit-learn
Operating System	Android 11+
Documentation	

Plagiarism Report

Abstract

Pakistani businesses often struggle with scattered data and expensive foreign software. Konvert is a management system with built-in AI that handles sales tracking and business operations. It is built for distribution companies and manages an eight-level hierarchy from sales representatives to the CEO, giving one unified data source.

By combining AI performance forecasting with real-time route tracking, Konvert automates routine sales work and cuts management overhead in one secure setup. The architecture scales from small businesses to large enterprises. The project keeps business data within Pakistan, replacing expensive foreign software with a local solution designed for national business management.

REVISION CHART

This chart contains a history of this document's revisions. The entries below are provided solely for illustration purposes. Those entries should be deleted until the revisions they refer to have actually been created.

The document itself should be stored in revision control, and a brief description of each version should be entered in the Revision Control System. A brief description can be repeated in this section. Revisions need not be described elsewhere in the document, unless they explain the document.

Version	Primary Author(s)	Description of Version	Date Completed
Draft	TBD	Initial draft created for distribution and review comments	(To be decided) TBD
Preliminary	TBD	Second draft incorporating initial review comments, distributed for final review	TBD
Final	TBD	First complete draft, which is placed under change control	TBD
Revision 1	TBD	Revised draft, revised according to the change control process and maintained under change control	TBD
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1 PROJECT INTRODUCTION AND CONTEXT

1.1 Core Motivations

The businessmen with good products have reached a digital dead end due to the fact that the sales management solutions which are available are either too costly or too difficult. This makes the business people spend more time on solving the technical issues rather than concentrating on sales. Using the foreign software for the purpose is also dangerous for the security of the data and the ownership rights. The problem which arises in the sales management is the lack of the local, secure, and easy-to-use solution where the business data can be stored.

1.2 Overview of the Proposed System

Konvert is an AI-driven sales management solution that aims to unite field work and executives, being the key automation solution for Pakistani businesses.

Going beyond the constraints of conventional management solutions, Konvert creates a state-of-the-art technical connection between field work on mobile devices and web-based platforms. Konvert is engineered to control and optimize an eight-layer organizational structure in order to make sure that information from the Sales Representatives at the street level is instantly available in the executives' dashboards. Integrating with the current web-based management systems, for example, Almashreqs infrastructure in case of Bristol, Konvert breaks the barriers of isolated data pools and creates a single source of truth.

The central component of the system is a forward-looking intelligence engine powered by machine learning and cutting-edge API that helps achieve ROI through live route tracking, automated tours, and AI-powered forecasting. Such predictive capabilities enable businesses to move away from reactive measures aimed at correcting their mistakes towards proactively expanding their operations and predicting future scalability and market trends. The system is scalable and suitable for use by enterprises of all sizes and automates sales-related processes while ensuring the security of national data.

1.3 Identification of the Problem

The dilemma of Pakistani companies is the choice between an out-of-date local solution and an expensive global software product. The utilization of obsolete foreign legacy software led to data inconsistencies, gaps in security, and lack of operational control. Companies such as Bristol are challenged by the combination of two issues. Regional platform SanSFE provides a poor user interface without synchronization with centralized systems resulting in data loss. Global software products such as Salesforce and HubSpot are too expensive and complicated for the local market.

The absence of synchronization between applications used in the field and centralized databases results in reporting errors that are more than 20 percent. Lack of alignment with the eight-level structure leads to operational inefficiencies and strategic mistakes made based on out-of-date data.

The main problems needing immediate solution are:

1. **Data Desynchronization:** The absence of two-way synchronization between the mobile application and the database results in mismatched records and redundant work.
2. **Operational Stagnation:** The lack of local AI-based optimization of the route leads to inefficient processes and resource wasting.
3. **Strategic Blindness:** Managers up to the CEOs lack real-time visibility and predictive forecasting, resulting in reactive management.

4. **Loss of Sovereignty:** Data concerning businesses in the nation being hosted on the foreign servers exposes businesses to privacy invasion and long-term security concerns.
5. **Economic Exclusion:** The cost of foreign subscriptions and learning curve deters local entrepreneurs and businesses from adopting advanced technology.

Without a proper system in place, the Pakistan companies will continue to make losses to foreign companies by working with inaccurate data.

1.4 Key Objectives and Goals

The purpose of our research is to develop the primary automation system for Pakistani businesses which would make their scattered field data work as one engine towards development. Some of our goals include:

1. **Establish Data Synchronization:** Design an integrated two-way system between field applications and web databases to provide a single source of truth throughout the organization.
2. **Automate Backbone Workflows:** Reduce management burdens with automated administrative sales processes, reporting process, and stock monitoring, which will allow them to concentrate on expanding.
3. **Optimize Operational Intelligence via AI:** Utilize Artificial Intelligence performance analysis and prediction capabilities for converting raw field data into predictive data.
4. **Maximize Business ROI through Localized Efficiency:** Increase ROI in Pakistan through provision of world-class route optimization and tours planning at reduced costs compared to foreign software.
5. **Safeguard National Data Sovereignty:** Design a safe local environment for managing your business and keeping your sensitive information in Pakistan.
6. **Catalyze Enterprise Scalability:** Design an elastic architecture to enable businesses to scale easily.

It is our mission to revolutionize the business process in Pakistan by transforming sales management from a tedious manual process to a proactive system driven by data. It is our commitment to offering high-quality technical solutions that are easy to use and cost-effective, thus enabling the local industry to grow and build momentum without any foreign technology.

2 DOMAIN AND STAKEHOLDER ANALYSIS

2.1 Target Customers and Users

Konvert’s customers split into two enterprise segments and three user types. Both shape everything that follows in this chapter.

2.1.1 Primary Enterprise Segments

Large-Scale Pharmaceutical and Distribution Enterprises: These companies run an eight-tier hierarchy, and the real problem is lag — field data can sit for days before it reaches anyone who can act on it. Konvert puts everyone in one environment instead, so executives see what’s happening in the field as it happens rather than reading about it a week later.

Small and Medium Enterprises (SMEs): SMEs make up most of Pakistan’s business landscape, but serious sales technology is usually out of their price range. Konvert is local, sized for them, and cheap enough to matter — and it automates enough of the routine work that they can grow without adding a whole new layer of admin staff just to keep up.

2.1.2 Key User Archetypes

Executive Leadership (CEO and C-Suite): Executives care about the big picture — national sales trends, regional performance, AI forecasts for planning expansion. Underneath all of that, what they’re actually protecting is market share and ROI.

Sales Managers (Tier 2–7): These are the people running teams day to day. Automated reporting and real-time visibility take some of the grind off their plate, so they can spot which routes are underperforming and coach with actual numbers instead of gut feel.

Field Sales Representatives (Tier 8): Reps produce most of the system’s data, and paperwork is their main complaint. The mobile interface takes care of route tracking, tour planning, and inventory, so more of the day is left for actually selling.

2.2 Stakeholder Mapping

Getting Konvert into the world takes a mix of internal and external stakeholders, and they don’t all want the same thing out of it.

2.2.1 Internal Project Stakeholders

Table 2.1: Summary of Internal Project Stakeholders

Stakeholder	Role in System
Project Development Team (Asim, Obaid, Farheen)	Handle the full-stack engineering, train the AI models, and build the two-way synchronization logic.
Project Advisor (Hafiz Ahsan Arshad)	Advises on strategy and validates the technical approach, keeping the project aligned with professional engineering standards and academic requirements.

2.2.2 External Project Stakeholders

Table 2.2: Summary of External Project Stakeholders

Stakeholder	Role in System
Bristol Mayer Biotech (Abdul Razzaq, Business Unit Head)	The lead pilot partner and main end-user group. They supply the operational context and serve as the primary test site for the eight-tier hierarchical visibility.
Almashreqs Soft. (Bashir Ahmed, CEO)	Manages the central web database. They provide the secure API endpoints and keep the web-to-mobile data bridge working correctly.
Local SME Sector	A secondary group that validates the market, giving feedback on how affordable and easy to adopt the system actually is for non-enterprise businesses.

2.3 Broader Social and Economic Impacts

Konvert's effects don't stop at the companies that buy it. They show up in how labor gets used and where money ends up staying.

Social Impact: Automating the clerical parts of the job means lower-tier field workers spend less time on data entry and more on actual selling — a small shift, but a real one, toward better mobility within the workforce. The interface doesn't ask much of anyone in terms of training, so people from very different educational backgrounds can pick it up without the usual dread that comes with learning enterprise software.

Economic Impact: Konvert is a homegrown alternative to CRM/ERP systems built and billed abroad, which keeps subscription money inside the country instead of flowing out. It also puts SMEs — the businesses actually doing most of the growing — on more even footing with large distributors, giving them the same caliber of tools at a price they can afford.

Organizational Impact: Inside a company, Konvert nudges the culture away from fixing things after they've already gone wrong and toward catching them early. Close the information gaps that plague multi-tier hierarchies and you get less friction between departments, lower overhead, and — eventually — a healthier bottom line.

2.4 System Dependencies and External Integrations

Konvert depends on a handful of external services to function at enterprise scale.

1. **Centralized Web Infrastructure (Almashreqs):** The existing Almashreqs web infrastructure is the legacy data source. Konvert uses custom API bridges to pull user hierarchies from it and push field activity logs back to the central Bristol servers.
2. **Geospatial Services (Google Maps API):** Google Maps Platform handles geocoding, live sales-man tracking, and route distance calculations — needed to confirm field attendance and plan tours efficiently.
3. **Machine Learning Frameworks:** Scikit-learn and TensorFlow train the predictive performance models, turning historical field data into forecasts.
4. **Google Firebase Ecosystem:** The Firebase service takes care of the back-end operations that happen in time. It uses Firestore to keep everything in sync when we are not using databases. The Firebase service also uses Firebase Cloud Messaging to send us push notifications. When it comes

to logging in, the Firebase service uses Firebase authentication to make sure we are who we say we are.

5. **Identity and Security Services:** Google Authenticator adds Multi-Factor Authentication (MFA) on top, securing executive access and sensitive commercial data.

2.5 Supporting Reference Documents

2.5.1 Review of Related Projects

To figure out where Konvert actually stands, we looked at what's already out there, regionally and globally.

SanSFE (Regional Industry Leader): SanSFE is the legacy baseline in South Asian pharmaceutical distribution. It does basic field reporting fine, but there's no real two-way sync — data coming from mobile usually has to be reconciled with the web servers by hand, and that's where the delays and errors in hierarchical reporting come from.

Salesforce and HubSpot (Global Enterprise CRM Solutions): These are the industry standard on paper, but they don't fit the Pakistani SME or distribution market. Subscription costs run in dollars, which is already a barrier, and their generic architecture almost always needs an expensive consultant to bend it into shape for local pharmaceutical or multi-tier distribution workflows.

Localized Legacy ERPs: A lot of local enterprises are still running custom or older ERP systems — desktop-bound or web-only, with little to no mobile support for field tracking. They're built to keep records, not to do anything with them: no geospatial tracking, no AI forecasting, no help planning routes.

2.5.2 Competitive Feature Comparison

Table 2.3: Comparative Analysis of Platform Features

Feature	SanSFE (Regional)	Global CRM (Salesforce)	Konvert (Proposed)
Data Sync	Manual reconciliation	High-speed cloud sync	Localized two-way flow
Mobile UI	Outdated legacy design	Modern but complex	Simplified field-force UX
Reporting	Static PDF logs	Advanced BI dashboards	Automated local metrics
Cost	Moderate local	High dollar-based	Affordable for local SME
Visibility	Immediate superior only	Customizable/Complex	Native 8-tier hierarchy
Intelligence	Record-based only	High-cost AI add-ons	Built-in native AI engine
Offline Work	Poor/Unstable	Basic/Cloud-dependent	Strong offline-first engine
Onboarding	Lengthy setup	Long consultant cycles	Agile rapid deployment
Support	Limited regional	Remote/Indirect	Direct local engineering

2.5.3 Reference Documents Overview

Technical Standards and Quality Framework:

- ISO/IEC 25010:2011 – System and software engineering quality models.
- ISO/IEC/IEEE 12207:2017 – Software life cycle processes.
- IEEE Std 830-1998 – IEEE Recommended Practices for Software Requirements Specifications (SRS).
- NIST Special Publication 800-63B – Digital Identity Guidelines.
- Human-centered design for interactive systems.
- IEEE 1012 – Standard for System, Software, and Hardware Verification and Validation (V&V).

Technical specifications and implementation:

- The specification for W3C Geolocation API.
- Software Engineering Guide to the Software Engineering Body of Knowledge (SWEBOK Guide V3.0).
- Fielding, Richard Twyman. (2000) – Architectural Styles and the Design of Network-based Software Architectures (REST).
- Google Maps Platform Technical Reference.
- Firebase RTL Architecture Patterns.

Research and Data Sovereignty:

- Hochreiter, S., & Schmidhuber, J. (1997) – Long Short-Term Memory (LSTM).
- General Data Privacy Principles (GDPR/APPI) – Legal frameworks ensuring data sovereignty, residency, and privacy.

3 SYSTEM REQUIREMENTS ANALYSIS

3.1 Core System Requirements

3.1.1 Detailed Functional Requirements

The functional requirements listed below outline what Konvert needs to do in order to assist with the day-to-day sales process, starting from logon until end-of-day reporting.

The FR Module 1 focuses on initializing, securing and accessing the system. The module 1 of FR is about initialising, securing and accessing the system.

- FR-1.1 Security and stability checks, as well as checks for updates, have to be performed by the app during the sequence of the splash screen before access is granted.
- FR-1.2 **Error Handling and Recovery:** Self healing should be provided in the system when an error occurs during runtime (cache clear, database resync, reinstall).
- FR-1.3 **Domain and Hierarchy Onboarding:** Users should be able to access a certain company domain (e.g. Bristol) through API keys and download the correct 8 tier hierarchy.
- FR-1.4 **Multi-Factor Authentication (MFA):** Login is secured using Google Authenticator tokens and Almashreqs' database hash.
- FR-1.5 **Feature Locking Mechanism:** The app must restrict feature access based on the user's role within the retrieved hierarchy.

FR Module 2: Data Synchronization and Persistence

- FR-2.1 **Master Sync Protocol:** On login, the system must run a full sync with the Almashreq cloud CRM, downloading only the data tied to that specific UserID.
- FR-2.2 **Bi-Directional Real-Time Sync:** Field data (orders, logs) must sync instantly with the Almashreq database, and web-side updates must reach the mobile app right away.
- FR-2.3 **Offline-First Architecture:** The app must stay usable without internet access, queueing transactions locally and uploading them automatically once the connection returns.

FR Module 3: Intelligent Dashboard and Analytics

- FR-3.1 **AI Performance Prediction:** The dashboard must show an AI-generated forecast of the user's weekly success probability, based on past behavioral patterns.
- FR-3.2 **Target Visualization:** Real-time graphs must show daily and weekly target progress against the active tour plan.
- FR-3.3 **Automated Notification System:** A background checker must fetch and surface prioritized alerts (warnings, updates, approvals) in a dedicated notification center.
- FR-3.4 **Advanced Reporting Module:** The system must produce detailed performance reports with granular filters (daily, weekly, categorical).

FR Module 4: Tour Planning and Roadmap

- FR-4.1 **Automated Tour Plan Creation:** Users must be able to create new tour plans by picking dates, routes, and expense budgets.

- FR-4.2 **AI Enhancement (“Enhance Button”):** The system must include an AI-helper class that optimizes the tour plan before submission.
- FR-4.3 **Management Approval Workflow:** Once submitted, tour plans must lock and stay locked until the management tier sends a digital approval.

FR Module 5: Fieldwork and Geospatial Operations

- FR-5.1 **Integrated Route Navigation:** The fieldwork interface must embed Google Maps so users can see the route to the next customer without leaving the app.
- FR-5.2 **Geo-Fencing Triggers:** Product selection buttons must stay inactive until the user’s GPS coordinates enter a defined radius around the customer’s location.
- FR-5.3 **Quick-Update Mechanism:** The fieldwork page must support quick, localized data fetches from the server without triggering a full Master Sync.

FR Module 6: Order Management and Booking

- FR-6.1 **Dynamic Product Selection:** The system must let users select OTC and Regular products, applying discounts and bonuses per the current logic.
- FR-6.2 **Daily Booking Summary:** Users must be able to review, edit, and confirm daily bookings before final submission.
- FR-6.3 **End-of-Day Upload:** The system must compile all daily bookings and upload them to the Al-mashreqs cloud CRM for administrative approval.

3.1.2 Detailed Non-Functional Requirements

The non-functional requirements below set out the quality bar, design limits, and performance targets the system has to meet.

NFR Module 1: Performance and Efficiency

- NFR-1.1 **Rendering and Sync Speed:** The Flutter UI must hold at least 15 FPS during navigation, and the initial Master Sync must finish within 15 seconds on a standard 4G connection.
- NFR-1.2 **Battery Optimization:** Background services – including 5-minute GPS polling and AI data gathering – must use less than 15% of device battery over an 8-hour shift.

NFR Module 2: Reliability and Availability

- NFR-2.1 **Offline Resilience:** The system must keep data loss during network dropouts close to zero, relying on strong local caching before it attempts to sync with Firebase.
- NFR-2.2 **High Availability:** The cloud infrastructure and Al-mashreqs API bridges must hold a 95% uptime SLA so executive monitoring and data flow aren’t interrupted.

NFR Module 3: Security and Compliance

- NFR-3.1 **Encryption Standards:** All data in transit should be encrypted via TLS 1.3 and all data on the local cache should be encrypted using AES-256 (Customer PII, bookings).
- NFR-3.2 **Breach Management:** The system must auto-expire idle sessions, wipe local data immediately, and alert admins if it detects a device security breach or root access.

NFR Module 4: Scalability and Maintainability

- NFR-4.1 **Concurrent Architecture:** The SQL structure and API bridges must handle 50+ simultaneous active field agents without hitting write-lock bottlenecks.

NFR-4.2 **Application Updates:** The system must support forced mandatory updates as well as optional delayed updates (1–2 days), so all tiers stay on the same compliant software version.

NFR Module 5: Usability and Compatibility

NFR-5.1 **Hardware Compatibility:** The application must run efficiently on lower-tier mobile devices (minimum 4GB RAM), since these are what most regional field representatives are issued.

NFR-5.2 **Human-Centered UX:** Based on ISO 9241-210, there are core field workflows that require no more than three clicks to access, or three taps on touchscreens, to avoid friction.

3.2 Identified System Actors

Konvert touches the following stakeholders:

- **Executive Leadership (CEO / Business Unit Head):** Top-tier decision makers who need national-level dashboards and AI forecasting.
- **Tactical Managers (National / Regional / Zonal / Area):** Mid-level supervisors managing teams across tiers 2–7, approving tour plans and tracking performance.
- **Operations Support (Command Center Operatives):** Backend staff who handle system alerts, data reconciliation, and technical support.
- **Field Sales Representatives (Tier 8):** Frontline staff running daily tours, placing orders, and generating field data.
- **System Administrator:** Handles domain configuration, user management, security policy, and system updates.

3.3 Comprehensive Catalog of System Use Cases

Table 3.1: Comprehensive Catalog of System Use Cases

UCID	Use Case Name	Brief Description
UC-1	System Startup and Security Check	Performs security, stability, and update checks during app launch before granting access.
UC-2	User Authentication and Domain Setup	Connects to company domain via API keys and retrieves the 8-tier hierarchy.
UC-3	Multi-Factor Authentication	Validates Google Authenticator tokens against the central database hash.
UC-4	Master Data Synchronization	Performs full bi-directional sync with Almashreqs cloud CRM upon login.
UC-5	Offline Data Management	Queues transactions locally during network loss and auto-uploads upon reconnection.
UC-6	AI Performance Forecasting	Displays AI-generated weekly success probability based on historical patterns.
UC-7	Target Visualization and Reporting	Shows real-time graphs of daily/weekly target achievement vs. tour plan.
UC-8	Notification Management	Fetches and displays prioritized alerts via the notification center.
UC-9	Tour Plan Creation	Allows users to create weekly tour plans with dates, routes, and budgets.
UC-10	AI Tour Plan Enhancement	Uses AI-helper class and Google Maps to optimize tour routes for efficiency.
UC-11	Tour Plan Approval Workflow	Locks submitted plans until management sends digital approval signals.
UC-12	Route Navigation and Geo-Fencing	Embeds Google Maps for navigation and enables product buttons only within customer radius.
UC-13	Customer Check-in and Order Booking	Handles product selection, discount logic, voice remarks, and photo proof capture.
UC-14	End-of-Day Data Upload	Compiles all daily bookings and pushes them securely to Almashreqs CRM.
UC-15	System Administration and User Management	Manages domain config, role assignments, security policies, and forced updates.

3.4 Visual Use Case Diagrams

Konvert has too many functional requirements spread across too many roles in the eight-tier hierarchy to fit into one readable use case diagram, especially once printed on A4. So the use case model is split into four sub-diagrams, one per functional module.

The four modules are:

- Authentication and Security Module

- Data Synchronization and Offline Management Module
- Tour Planning and Field Operations Module
- Dashboard, Analytics, and Reporting Module

Each sub-diagram is described below, along with how its use cases relate to one another and to the actors involved.

3.4.1 Module 1: Authentication and Security

This module handles secure access to Konvert: checking device integrity, authenticating users, enforcing security policy, and managing accounts. The System Administrator oversees platform security and account management, while Field Sales Representatives and Tactical Managers log in through the domain-based authentication flow.

The flow starts with System Startup and Security Check, which verifies application integrity, checks for root access, and confirms version compliance. From there, users authenticate through User Authentication and Domain Setup. When MFA is turned on, that flow pulls in Multi-Factor Authentication as an extra step.

System Administration and User Management covers creating, modifying, disabling, and monitoring user accounts across all tiers.

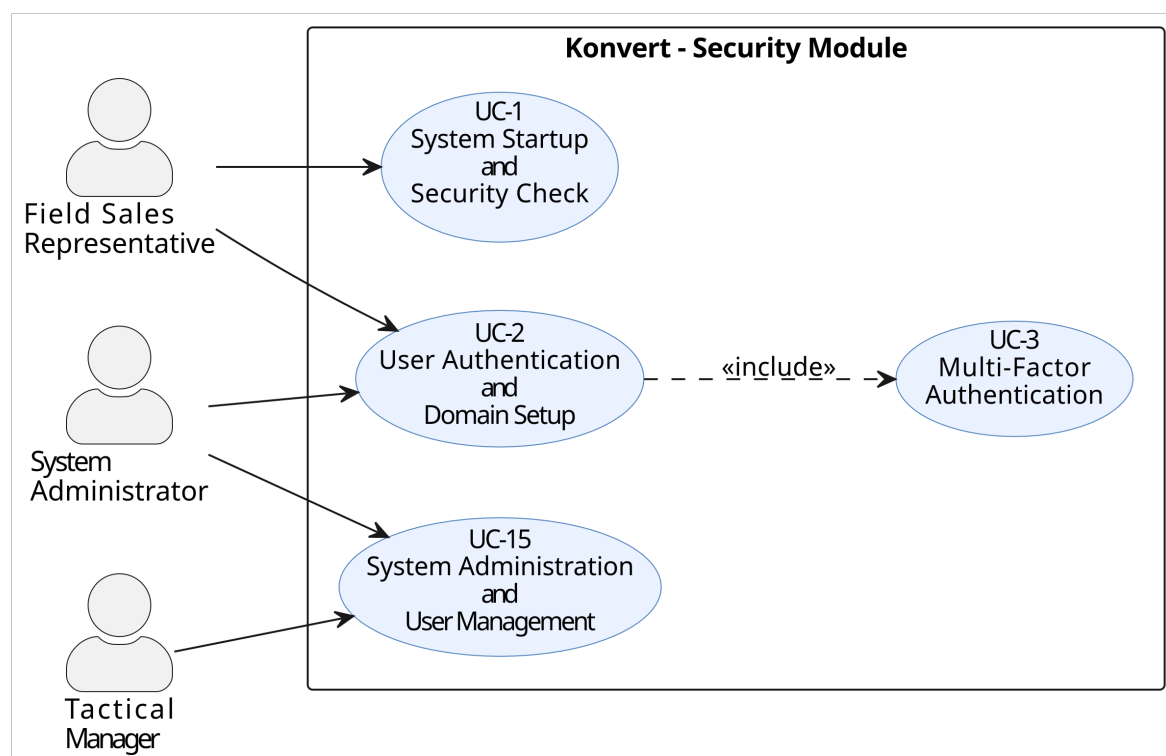


Figure 3.1: Visual Use Case Model for Authentication & Security

Figure 3.1 Description:

The diagram shows the System Administrator actor connected to:

- UC-1: System Startup and Security Check
- UC-15: System Administration and User Management

The Field Sales Representative and Tactical Manager actors are connected to:

- UC-2: User Authentication and Domain Setup

UC-2 includes UC-3 (Multi-Factor Authentication) through an «include» relationship. All use cases sit inside the system boundary labeled **Konvert – Security Module**.

3.4.2 Module 2: Data Synchronization and Offline Tracking

This module keeps operational data in sync between mobile devices and the Almasreqs cloud, whether or not the device has a connection at any given moment.

Master Data Synchronization uploads pending offline transactions and pulls down the latest customer, product, target, and tour plan data. Offline Data Management handles local storage, sync queues, and conflict resolution once connectivity comes back.

Operations Support keeps an eye on sync status and chases down failed transfers. The Almasreqs External System provides the backend services that the sync depends on.

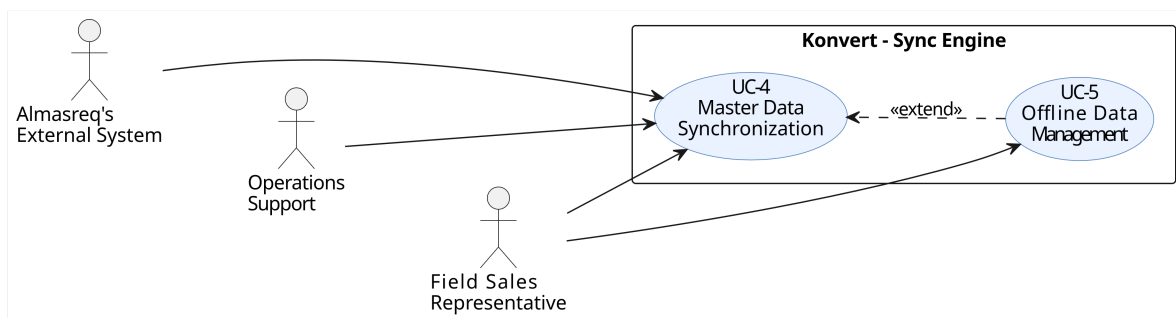


Figure 3.2: Visual Use Case Model for Data Synchronization

Figure 3.2 Description:

The diagram contains the following actor interactions:

- Field Sales Representative → UC-4 Master Data Synchronization
- Field Sales Representative → UC-5 Offline Data Management
- Operations Support → UC-4 Master Data Synchronization
- Almasreqs External System → UC-4 through API integration

UC-4 extends UC-5 under the condition *network unavailable*. All use cases sit inside the system boundary labeled **Konvert – Sync Engine**.

3.4.3 Module 3: Tour Planning and Field Operations

This module covers the core work a field rep does: build a tour plan, get the route optimized, navigate to each stop, check in with a geo-fence, and record the order.

Tour Plan Creation lets reps build a weekly visit schedule. From there they can optionally run AI Tour Plan Enhancement to have the route optimized. Once approved, that plan drives the day's field activity. Route Navigation and Geo-Fencing makes sure a rep is physically at a customers location before the order-booking screen opens up. Once the geo-fence check passes, Customer Check-in and Order Booking becomes available.

Tactical Managers review and approve submitted tour plans.

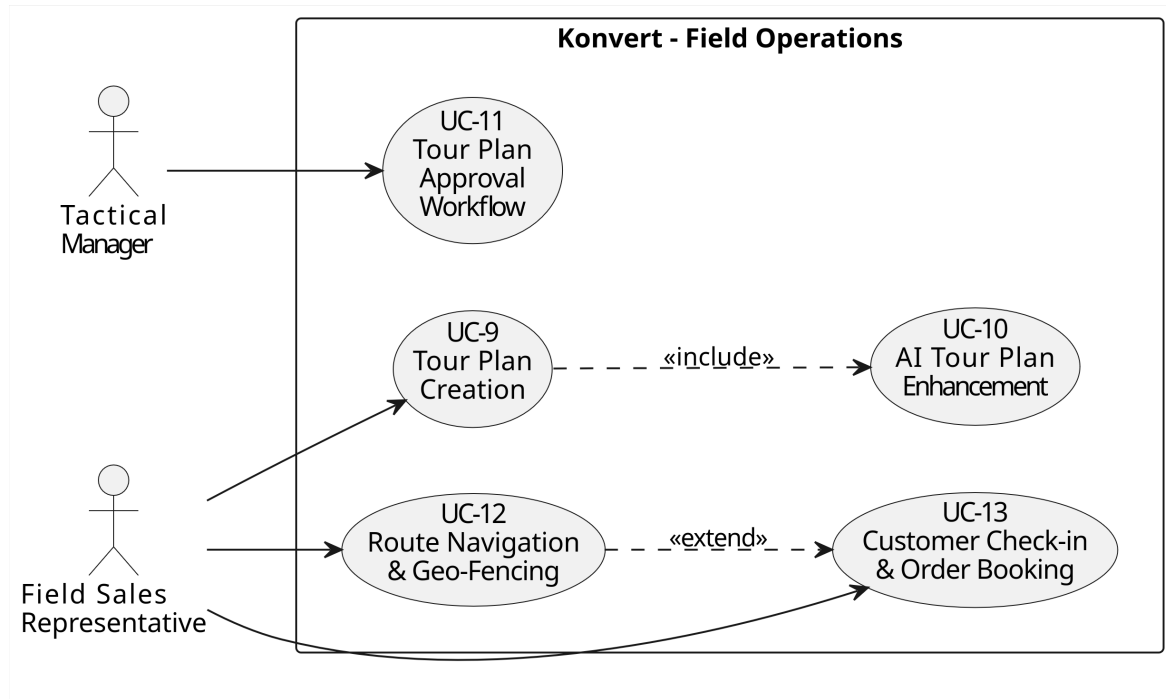


Figure 3.3: Visual Use Case Model for Tour Planning & Operations

Figure 3.3 Description:

The diagram contains the following actor interactions:

- Field Sales Representative → UC-9 Tour Plan Creation
- Field Sales Representative → UC-12 Route Navigation and Geo-Fencing
- Field Sales Representative → UC-13 Customer Check-in and Order Booking
- Tactical Manager → UC-11 Tour Plan Approval Workflow

UC-9 includes UC-10 AI Tour Plan Enhancement under the condition *user clicks enhance*. UC-12 extends UC-13 under the condition *GPS enters geo-fence*. All use cases sit inside the system boundary labeled **Konvert – Field Operations**.

3.4.4 Module 4: Intelligence Dashboard and Reporting

This module gives management the numbers: sales performance, target tracking, alerts, and forecasts. AI Performance Forecasting runs machine learning over historical data to predict outcomes and flag trends. Target Visualization and Reporting handles the graphical dashboards, KPI tracking, and side-by-side performance comparisons. Notification Management pushes alerts, reminders, and operational updates out to the org.

Field Sales Representatives use the reporting screens to check their own targets and numbers, while Tactical Managers and Executive Leadership have a broad perspective of the organization.

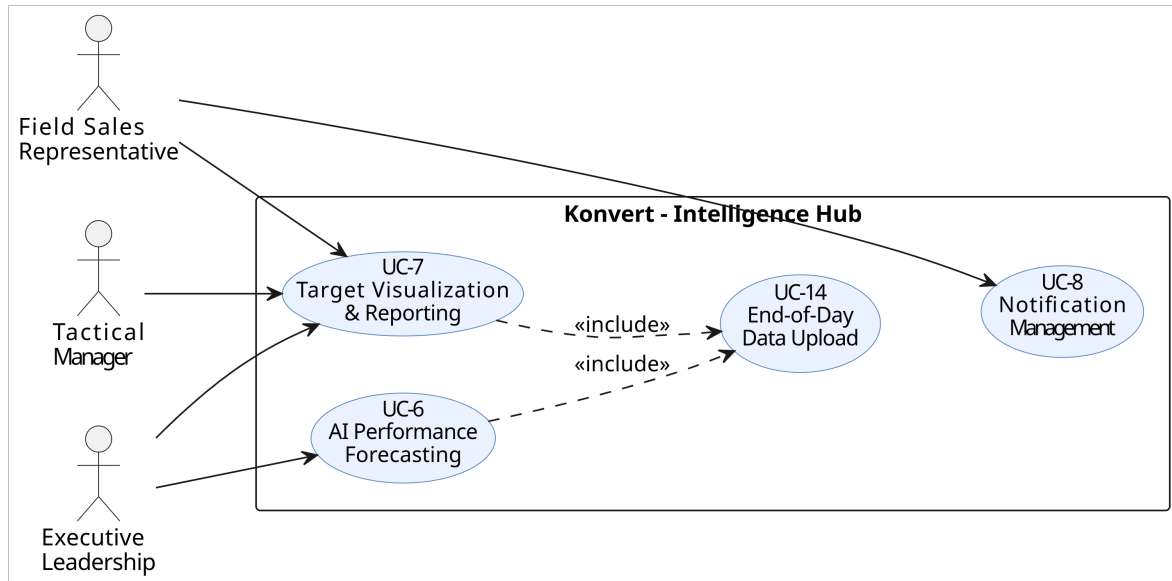


Figure 3.4: Visual Use Case Model for Dashboard & Intelligence Analytics

Figure 3.4 Description:

The diagram contains the following actor interactions:

- Executive Leadership → UC-6 AI Performance Forecasting
- Executive Leadership → UC-7 Target Visualization and Reporting
- Tactical Manager → UC-7 Target Visualization and Reporting
- Tactical Manager → UC-8 Notification Management
- Field Sales Representative → UC-7 Target Visualization and Reporting

Both UC-6 and UC-7 include UC-14 End-of-Day Data Upload as a prerequisite, through «include» relationships. All use cases sit inside the system boundary labeled **Konvert – Intelligence Hub**. Put together, these four diagrams show how security, sync, field operations, and reporting fit together across Konvert.

3.5 In-Depth Use Case Narratives

3.5.1 UC-2: User Authentication and Domain Setup

Table 3.2: Detailed Flow for UC-2: User Authentication and Domain Setup

Use Case ID:	UC-2
Use Case Name:	User Authentication and Domain Setup
Actors:	Primary: Field Sales Representative, Tactical Manager. Secondary: System Administrator, Al-mashreqs External System.
Description:	Lets a user connect to their company’s domain, checks their credentials, and pulls down the 8-tier hierarchy before the app grants access.
Trigger:	User launches the Konvert application after the splash screen security checks pass.
Preconditions:	1. The device has passed root/breach detection. 2. The app version is current (or update is accepted). 3. Internet connectivity is available.

Postconditions:	1. User session is established with a valid JWT token. 2. The 8-tier hierarchy is cached locally. 3. Role-based feature locks are applied.
Normal Flow:	1. System displays the Domain Configuration screen. 2. User enters the company API key (e.g., Bristol domain key). 3. System validates the key format locally. 4. System sends the key to the Almashreqs API bridge for verification. 5. Almashreqs returns the company configuration and 8-tier hierarchy. 6. System caches the hierarchy in local encrypted storage. 7. System displays the Login screen. 8. User enters username and password. 9. System hashes the password and sends it to Almashreqs for validation. 10. On success, system checks if MFA is enabled for the user. 11. If MFA is enabled, system triggers UC-3 (Multi-Factor Authentication). 12. Upon MFA success (or if skipped), system applies feature locks based on hierarchy tier. 13. System initiates UC-4 (Master Data Synchronization). 14. User is routed to the Dashboard.
Alternative Flows:	4a. Invalid API Key: If the key fails format validation, system shows “Invalid domain key” and returns to step 2. 5a. Network Error: If Almashreqs is unreachable, system checks for cached domain data. If present, it proceeds to step 7 in offline mode. If absent, it shows “Domain verification failed” and halts. 9a. Invalid Credentials: If password validation fails, system increments a failure counter. After 3 failures, the account is temporarily locked and an alert is sent to the System Administrator. 10a. First Login: If the user has no MFA setup, the system prompts the user to scan a QR code for Google Authenticator enrollment before proceeding.
Exceptions:	E1. Almashreqs API Timeout: System retries twice with exponential backoff. If all fail, it enters offline mode using last-known hierarchy data and queues a sync retry. E2. Corrupted Local Cache: If the hierarchy cache is corrupted, system clears storage and restarts at step 1, logging the error to the Security Manager.
Includes:	UC-3 (Multi-Factor Authentication), UC-4 (Master Data Synchronization)
Frequency of Use:	Once per app session (typically daily per user).
Special Requirements:	TLS 1.3 must be active for all API calls. Local cache must be AES-256 encrypted.

3.5.2 UC-4: Master Data Synchronization

Table 3.3: Detailed Flow for UC-4: Master Data Synchronization

Use Case ID	UC-4
Use Case Name	Master Data Synchronization
Actors	Primary: Field Sales Representative. Secondary: Operations Support, Almashreqs External System.
Description	Runs a full two-way sync between the mobile device and the Almashreqs cloud CRM: the device gets the latest data for the logged-in UserID, and any transactions saved offline get pushed up.
Trigger	Triggered automatically after successful login (UC-2) or manually via the Dashboard “Master Sync” button.
Preconditions	1. User is authenticated with a valid session. 2. Device has internet connectivity or valid offline queue data. 3. API bridge endpoints are responsive.
Postconditions	1. Local SQLite database is updated with latest doctors, chemists, products, and targets. 2. All queued offline bookings are uploaded to Almashreqs. 3. Sync timestamp is recorded. 4. Session is reinstated.

Normal Flow	1. System temporarily revokes active session status to prevent mid-sync actions. 2. System reads the local offline transfer queue from SQLite. 3. If queued items exist, system establishes a 3-way handshake with Almashreqs. 4. System uploads queued bookings, logs, and GPS traces in batch packets. 5. Almashreqs validates each packet and returns ACK codes. 6. System clears successfully uploaded items from the local queue. 7. System sends a download request for data assigned to the current UserID. 8. Almashreqs returns doctors, chemists, stockists, products, targets, and tour plans. 9. System writes the downloaded data to local SQLite with AES-256 encryption. 10. System updates the Firebase Firestore mirror for real-time sync. 11. System reinstates the active session. 12. System shows the Dashboard with updated data.
Alternative Flows	3a. Empty Queue: If no offline items exist, system skips steps 3–6 and proceeds directly to step 7. 5a. Partial Upload Failure: If some packets fail validation, system retries those specific packets up to three times. Persistent failures are flagged for Operations Support review. 8a. No New Data: If Almashreqs reports no changes since the last sync, system updates only the sync timestamp and proceeds to step 11.
Exceptions	E1. Fatal Sync Failure: If the download phase fails after upload success, system keeps the session revoked, displays error code SYNC-DL-01, and offers Retry or Logout. E2. Database Write Lock: If SQLite encounters a write lock during step 9, the system pauses for 500 ms and retries up to five times before aborting and alerting the user.
Includes	None
Frequency of Use	Once per login and on demand (average 2–3 times per day).
Special Requirements	Sync must complete within 15 seconds on a 4G connection. Packet size must not exceed 2 MB per batch.

3.5.3 UC-9: Tour Plan Creation

Table 3.4: Detailed Flow for UC-9: Tour Plan Creation

Use Case ID	UC-9
Use Case Name	Tour Plan Creation
Actors	Primary: Field Sales Representative. Secondary: Tactical Manager.
Description	A field rep builds a weekly tour plan here: pick working days, assign bricks/territories, choose customers, set an expense budget, and submit it to management for approval.
Trigger	User selects “Create Tour Plan” from the Dashboard or Tour Plan screen.
Preconditions	1. User is authenticated and has Tier 8 (Field Rep) role. 2. Master Sync has completed at least once. 3. Local customer and brick data is available.
Postconditions	1. A new tour plan record is created in local SQLite with status “Draft”. 2. If submitted, status changes to “Pending” and is locked from further edits. 3. A notification is queued for the assigned Tactical Manager.

Normal Flow	1. System displays the Tour Plan creation interface with a weekly calendar grid. 2. User marks each day as “Field Work” or “Holiday”. 3. For each Field Work day, user selects a brick/territory from the cached list. 4. System loads the customers (doctors/chemists) assigned to that brick. 5. User selects customers to visit and sets a scheduled time for each. 6. User enters the daily expense budget (fuel, food, miscellaneous). 7. System calculates total weekly mileage using Google Maps Distance Matrix API. 8. User reviews the complete plan summary. 9. User clicks “Submit for Approval”. 10. System locks the plan, sets status to “Pending”, and pushes a notification to the Tactical Manager via FCM. 11. System queues the plan for upload to Almashreqs on the next synchronization.
Alternative Flows	7a. AI Enhancement Requested: If the user clicks “Enhance” at step 7, the system triggers UC-10 (AI Tour Plan Enhancement) and returns to step 8 with the optimized route. 9a. Save as Draft: If the user clicks “Save Draft”, the system stores the plan with status “Draft” and allows future editing. No notification is sent. 10a. Offline Submission: If internet connectivity is unavailable, the system stores the plan locally with status “Pending” and automatically submits it when connectivity is restored.
Exceptions	E1. No Customers in Brick: If a selected brick has no assigned customers, the system displays a warning and returns to step 3. E2. Budget Exceeds Limit: If the entered budget exceeds the company-defined limit for the user’s tier, the system displays an error and requires adjustment before proceeding.
Includes	UC-10 (AI Tour Plan Enhancement) – Optional
Frequency of Use	Weekly (typically once per week per field representative).
Special Requirements	The plan should be creatable within 10 screen taps. Google Maps Distance Matrix API requests must be batched to minimize operational cost.

3.5.4 UC-12: Route Navigation and Geo-Fencing

Table 3.5: Detailed Flow for UC-12: Route Navigation and Geo-Fencing

Use Case ID	UC-12
Use Case Name	Route Navigation and Geo-Fencing
Actors	Primary: Field Sales Representative. Secondary: Google Maps External System.
Description	Embeds turn-by-turn navigation to the next customer and locks product selection and order booking until the rep’s GPS position is actually inside the customer’s geo-fence radius.
Trigger	User selects a customer from the active tour plan and clicks “Maps” or “Check In”.
Preconditions	1. User has an approved tour plan for the current day. 2. GPS is enabled on the device. 3. Customer location data (latitude and longitude) is available locally.
Postconditions	1. User is guided to the customer location through embedded Google Maps. 2. Upon entering the geo-fence radius, product selection buttons are enabled. 3. Check-in timestamp and GPS coordinates are logged locally.

Normal Flow	1. System displays the day's tour plan with a list of scheduled customers. 2. User selects the next customer and clicks "Maps". 3. System launches the embedded Google Maps view with the route displayed. 4. System polls GPS every five minutes to track movement. 5. As the user approaches the destination, the system calculates the distance to the customer coordinates. 6. When the distance falls below the geo-fence radius (default 50 meters), the system displays a local notification stating: "You have entered the customer zone." 7. System enables the "Select Products" and "Place Order" buttons. 8. User clicks "Check In" and the system records the timestamp and exact GPS coordinates. 9. System transitions to UC-13 (Customer Check-in and Order Booking).
Alternative Flows	3a. External Maps Preferred: If the user selects "Open in Google Maps App", the system passes the customer coordinates to the external Google Maps application and continues GPS polling in the background. 6a. GPS Signal Lost: If GPS accuracy drops below 20 meters, the system falls back to network-based location services. If both methods fail, the user is required to capture a location-proof photograph before proceeding. 7a. Early Arrival: If the user arrives before the scheduled visit time, the system allows check-in but displays a warning message indicating an early arrival.
Exceptions	E1. GPS Spoofing Detected: If the Security Manager detects mock-location software or GPS spoofing attempts, the system disables all fieldwork features, terminates the current session, and alerts the System Administrator. E2. Customer Location Mismatch: If the user's location remains more than 500 meters away from the stored customer coordinates for over 10 minutes, the system flags a potential address error and prompts the user to request a location update subject to manager approval.
Includes	UC-13 (Customer Check-in and Order Booking)
Frequency of Use	Approximately 10–20 times per day per field representative.
Special Requirements	GPS polling must not exceed 15% battery consumption over an 8-hour work-day. Geo-fence calculations must complete within 2 seconds.

3.5.5 UC-13: Customer Check-in and Order Booking

Table 3.6: Detailed Flow for UC-13: Customer Check-in and Order Booking

Use Case ID	UC-13
Use Case Name	Customer Check-in and Order Booking
Actors	Primary: Field Sales Representative.
Description	Covers the actual customer visit: the rep selects products, applies discounts and bonuses, adds a voice or text remark, snaps a photo as proof, and locks in the booking for the end-of-day upload.
Trigger	User completes UC-12 geo-fence check-in and the system enables the booking interface.
Preconditions	1. Geo-fence validation has been passed (or photo fallback is approved). 2. Product catalog has been synchronized locally. 3. User has an active approved tour plan.
Postconditions	1. A booking record is created in local SQLite with status "Queued". 2. Inventory counts are adjusted locally. 3. Daily booking summary is updated.

Normal Flow	1. System displays the customer profile and booking interface. 2. User selects either the “OTC” or “Regular” product category. 3. System filters the product catalog accordingly. 4. User selects products and enters quantities. 5. System automatically applies current discount and bonus rules. 6. User reviews the order line items and total amount. 7. User taps the microphone icon to record a voice remark in Urdu or English. 8. System transcribes the audio locally using on-device AI and stores both the audio and text versions. 9. System requests a mandatory photograph of the customer premises or available stock. 10. User captures the photograph and the system watermarks it with GPS coordinates and a timestamp. 11. User clicks “Finalize Booking”. 12. System stores the complete booking in local SQLite with status “Queued”. 13. System updates the tour plan visit status to “Completed”. 14. System returns to the tour plan list and displays the next scheduled customer.
Alternative Flows	5a. Manual Discount Override: If the user has manager-approved override rights, the system permits manual discount entry beyond automatic rules and requires a digital signature or approval code. 7a. Text Remark Only: If the user skips voice recording, the system allows direct text entry in Urdu or English. 9a. Photo Fallback for GPS Issues: If geo-fencing was bypassed because of GPS failure, the captured photograph is flagged as “Location Proof” and requires manual verification by backend staff. 11a. Save as Pending: User may save the booking with status “Pending” and edit it later before finalization.
Exceptions	E1. Product Out of Stock: If a selected product has zero available stock, the system blocks the item and suggests alternative products from the same category. E2. Local Database Write Failure: If SQLite storage fails because of corruption or write errors, the system exports the booking as a JSON file to local device storage and prompts the user to initiate database repair.
Includes	None
Frequency of Use	Approximately 10–30 times per day per field representative.
Special Requirements	The core booking workflow should complete within three screen taps after product selection. Photo watermarking must be tamper-evident and verifiable.

3.5.6 UC-14: End-of-Day Data Upload

Table 3.7: Detailed Flow for UC-14: End-of-Day Data Upload

Use Case ID	UC-14
Use Case Name	End-of-Day Data Upload
Actors	Primary: Field Sales Representative. Secondary: Almashreqs External System, Operations Support.
Description	Packages the day’s bookings, logs, GPS traces, and remarks into one payload and pushes it to the Almashreqs cloud CRM for admin processing and inventory adjustment.
Trigger	User clicks “End Day” on the Dashboard, or the system automatically triggers the upload at 18:00 if a manual upload has not been performed.

Preconditions	1. At least one booking exists in local SQLite with status “Queued”. 2. User session is valid. 3. Almashreqs API bridge is reachable or data is marked for delayed synchronization.
Postconditions	1. All queued bookings are uploaded and their status changes to “Uploaded”. 2. Local daily summary is archived. 3. Almashreqs inventory is updated. 4. User receives a confirmation notification.
Normal Flow	1. System displays the Daily Booking Summary screen containing all queued bookings. 2. User reviews each booking for accuracy. 3. User clicks “Confirm and Upload”. 4. System compiles bookings, GPS traces, photographs, and remarks into a compressed payload. 5. System encrypts the payload using AES-256. 6. System establishes a TLS 1.3 connection to the Almashreqs upload endpoint. 7. System transmits the payload using chunked packets and retry logic. 8. Almashreqs validates data integrity and returns a batch receipt identifier. 9. System updates local booking statuses to “Uploaded”. 10. System clears the daily queue and archives the summary. 11. System sends a success notification to the user and the assigned manager. 12. System records the upload event in the audit trail.
Alternative Flows	2a. Edit Before Upload: User may select any booking and edit or delete it before final submission. 3a. Offline Auto-Queue: If internet connectivity is unavailable, the system stores the compiled payload in a “Delayed Upload” queue and automatically uploads it when connectivity is restored. 8a. Partial Validation Failure: If Almashreqs rejects specific bookings, such as duplicate records, the system marks those bookings as “Failed”, displays the corresponding error codes, and allows individual retry operations.
Exceptions	E1. Upload Timeout: If the upload process exceeds 60 seconds, the system pauses the transfer, retries using smaller packet sizes, and informs the user of the delayed completion. E2. Authentication Expiry Mid-Upload: If the session token expires during transmission, the system refreshes the token through Firebase Authentication and resumes the upload without data loss.
Includes	None
Frequency of Use	Once per day per field representative.
Special Requirements	Upload operations must resume automatically after interruption without data corruption. All photographs must be compressed to less than 500 KB before transmission.

4 DATA FLOW AND PROCESS DIAGRAMS

This chapter is about the Data Flow Diagrams for the Konvert system. It shows how data moves between users, other systems what happens inside the system and where the data is stored. We are going to look at this in three levels of detail.

4.1 Level 0 Context Diagram (System Overview)

We start with a picture view of the Konvert system. It is like a map that shows what is going in and out of the system. We see the users, the people in charge the leaders and the other systems that the Konvert system talks to.

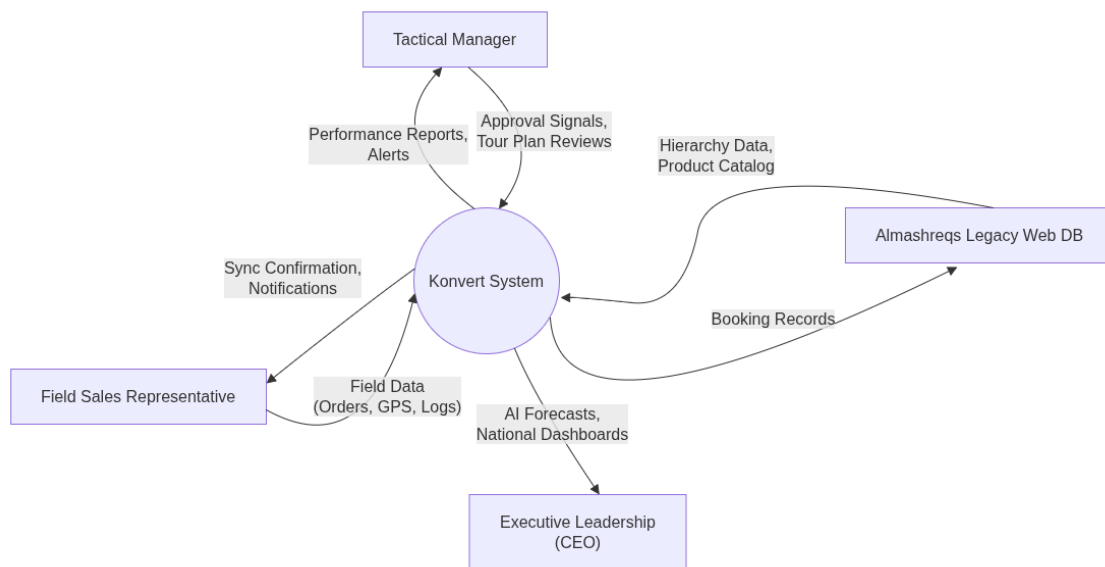


Figure 4.1: Level 0 Context Diagram (System Overview)

4.2 Level 1 Data Flow Diagram (Main Modules)

Zooming in one level, The Konvert system is made up of a main parts. These parts are how users log in how data is synchronized how tours are planned what happens in the field. How reports are made. This diagram shows how these parts work with the users, databases and other services.

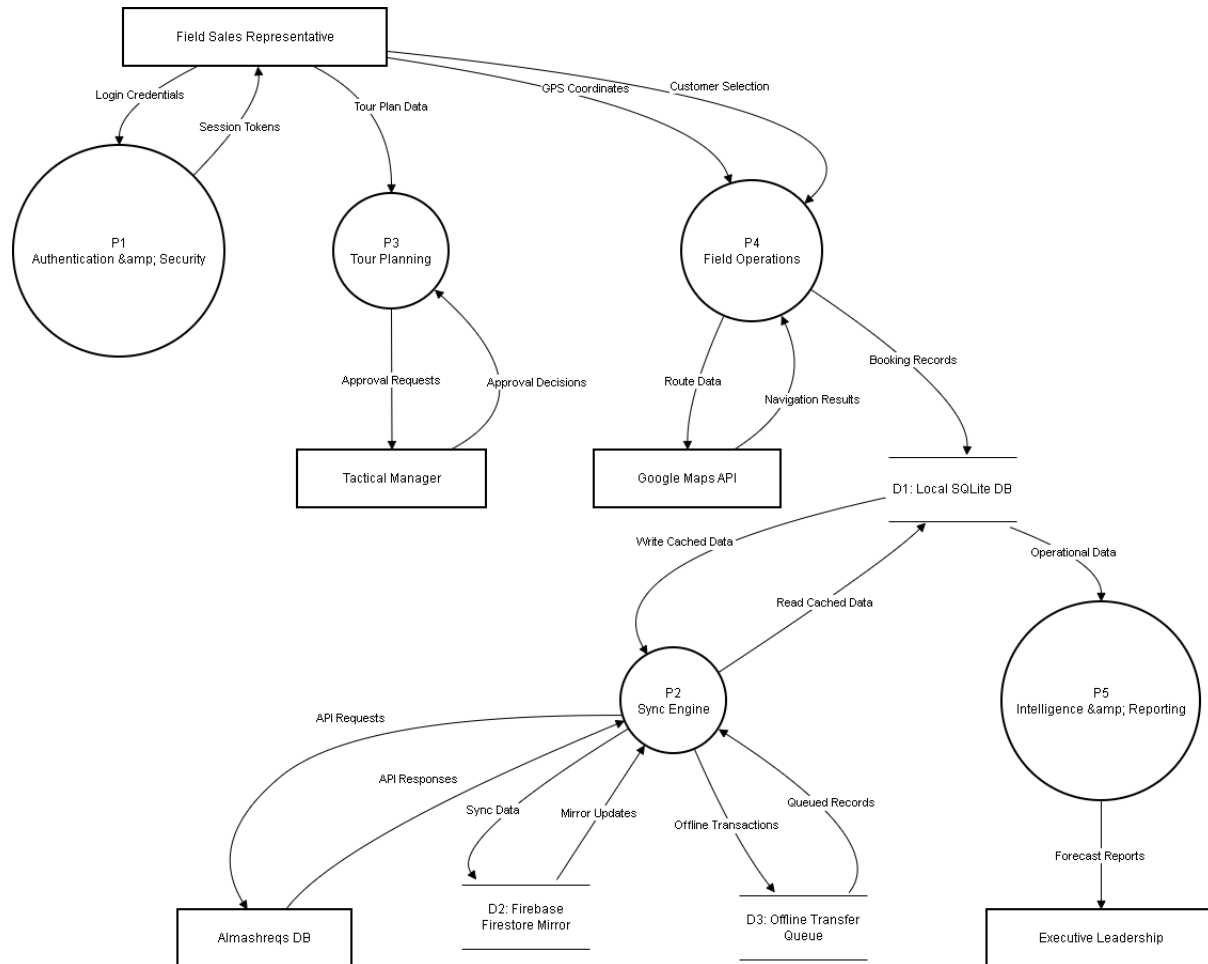


Figure 4.2: Level 1 Data Flow Diagram (Main Modules)

4.3 Level 2 Data Flow Diagram (Field Operations Details)

This diagram zooms into just the Field Operations part of the Konvert system. We will see how the system checks the GPS sets boundaries processes orders and captures media.

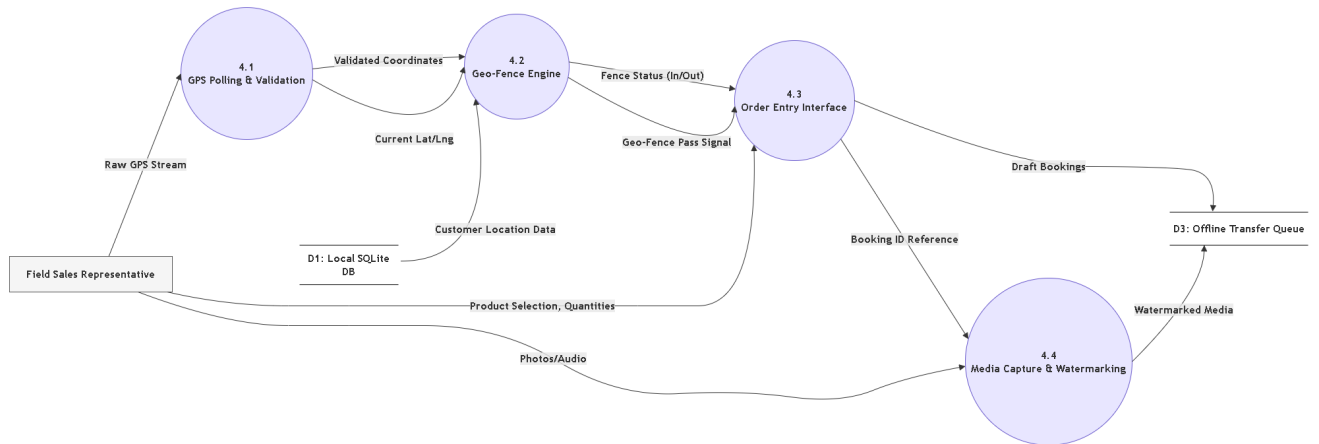


Figure 4.3: Level 2 Data Flow Diagram (Field Operations Details)

5 ARCHITECTURAL AND SYSTEM DESIGN

This chapter covers the structural and behavioral side of Konvert: the architecture, the components, how they talk to each other, and the database structure behind the whole thing. The diagrams below show how the various modules fit together.

5.1 System Architecture Visualization

Konvert is built in layers, separating the presentation layer, business logic, external services, and enterprise integration from one another. That split keeps the system easier to maintain and scale, and keeps the security boundary with the existing business infrastructure clean.

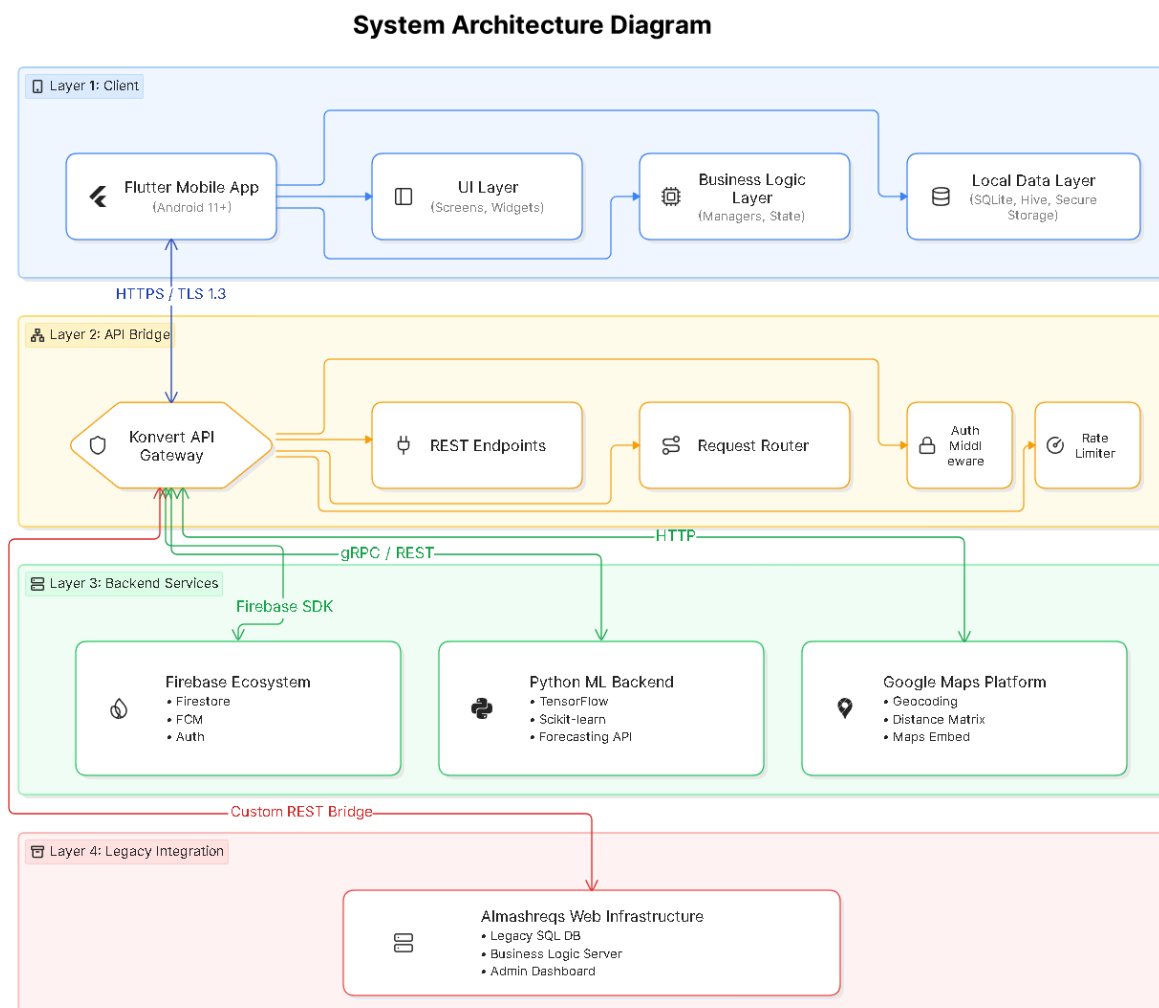


Figure 5.1: High-Level Architecture Visualization of Konvert

There are four layers. At the bottom sits the Flutter mobile app, which runs on Android and handles the UI, business logic, and local storage. Above that is an API gateway, responsible for routing requests, authentication, authorization, and general communication management. The third layer covers the external services Konvert depends on: Firebase, the Python-based machine learning services, and the Google

Maps APIs. The top layer is where the mobile ecosystem connects into the existing Almarshreqs enterprise infrastructure. Everything between layers runs over HTTPS/TLS with standard authentication.

5.2 Comprehensive Class Diagram

The class diagram lays out the static structure of Konvert – the major classes, what each one is responsible for, and how they relate to each other.

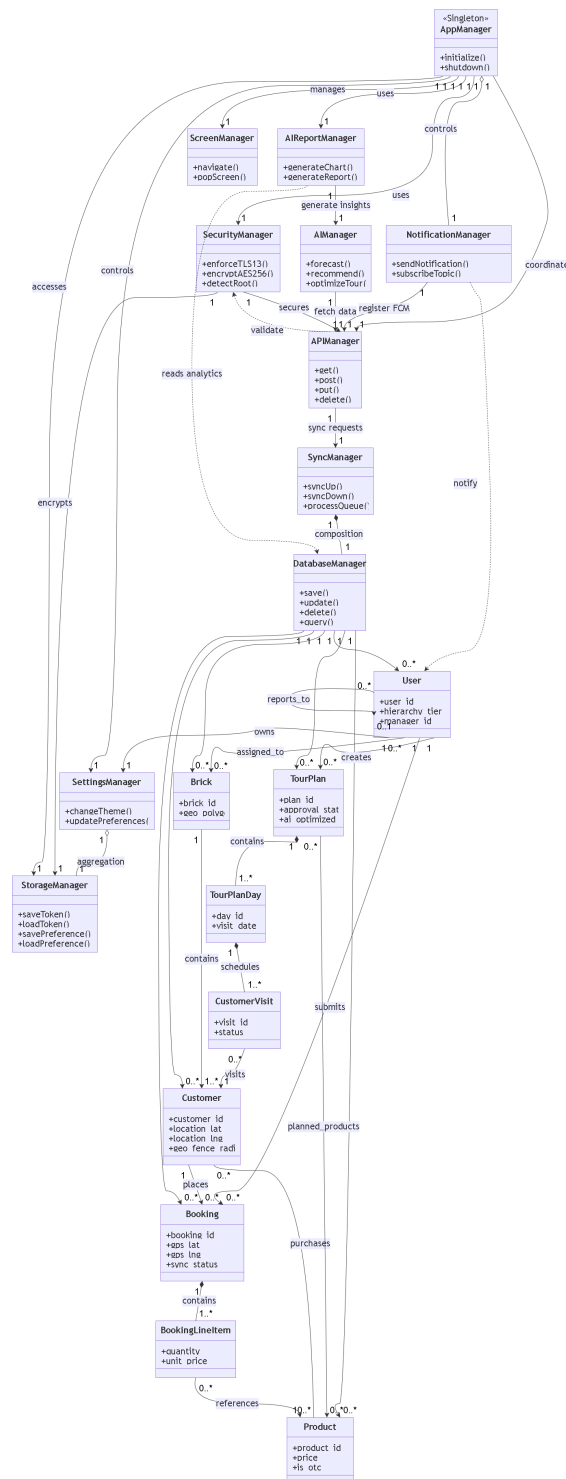


Figure 5.2: Comprehensive Class Diagram of the Konvert Ecosystem

AppManager sits at the center and coordinates communication across the major subsystems. ScreenManager handles navigation and screen transitions. APIManager manages REST calls to the backend. SyncManager takes care of syncing local and remote data. DatabaseManager handles persistence through SQLite and Hive. StorageManager keeps user credentials and app settings secure. SecurityManager enforces encryption, authentication, and device security checks. AIRportManager builds the forecasts and analytical reports. NotificationManager pushes notifications through Firebase Cloud Messaging. Everything else in the diagram is a business entity – users, customers, products, bookings, tour plans – rather than a manager class. Splitting responsibilities this way keeps each class focused on one job, which makes the codebase easier to maintain.

5.3 Collaboration Protocol: Master Synchronization Process

This diagram shows how the app’s components interact during a sync between the mobile app and the enterprise backend.

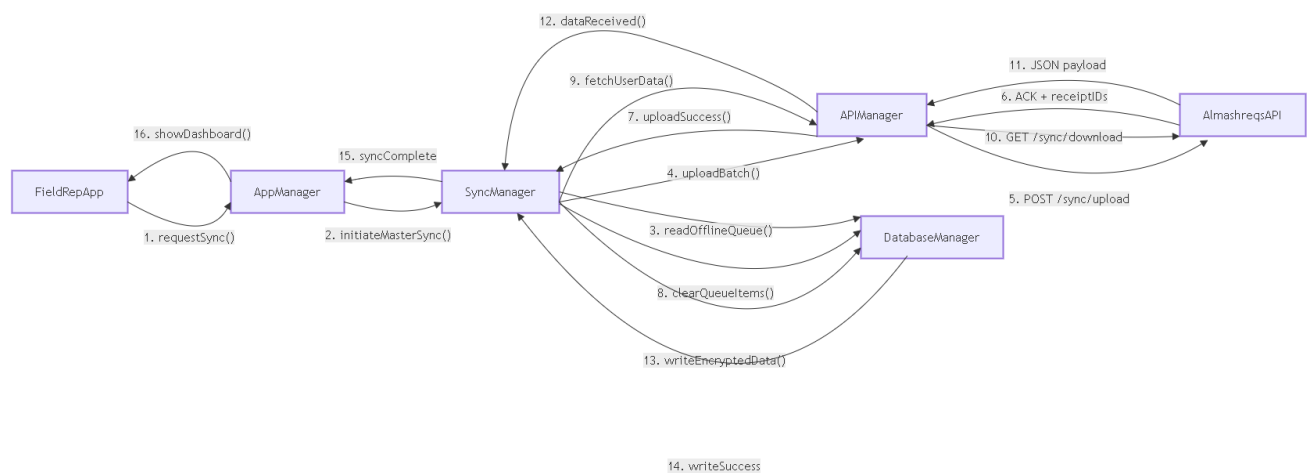


Figure 5.3: Collaboration Protocol for Master Data Synchronization

A field rep kicks off a sync from the app. AppManager passes the request to SyncManager, which pulls the offline records stored locally. Those records go up to the backend through APIManager. Once the upload is confirmed, the processed entries clear out of the local queue. From there, the system pulls down the latest organizational data, customer records, products, and tour plans. Finally, the local database updates and the dashboard refreshes with the new information – so the mobile device and the central servers stay in step.

5.4 Collaboration Protocol: Geo-Fence Validated Booking

This diagram covers how order booking works alongside the geo-fence check.

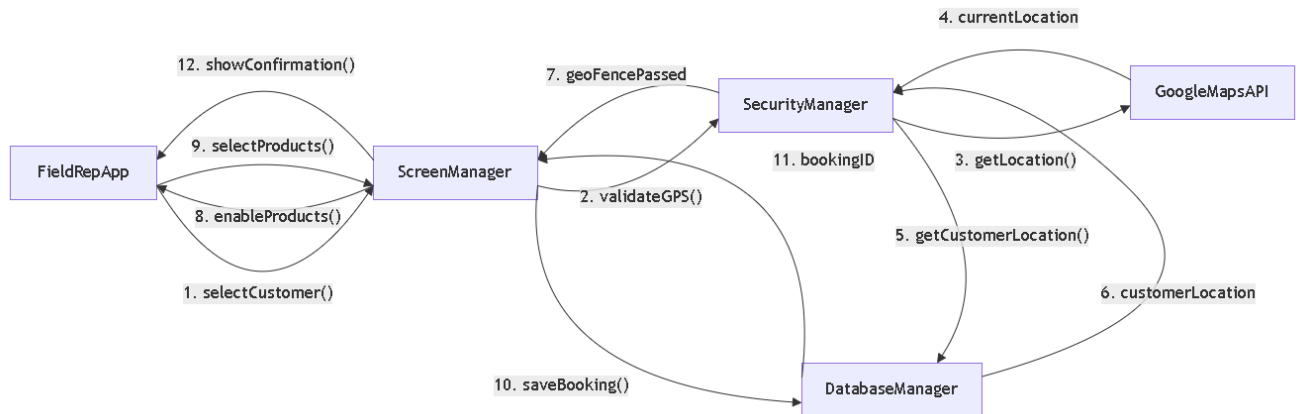


Figure 5.4: Collaboration Protocol for Geo-Fence Validated Booking

A field rep selects a customer, and the app pulls the current GPS position from Google Maps and checks it against that customer's registered location. If the rep is within the geo-fence radius, product selection and order entry open up. The booking is stored locally and queued for sync, and the user gets a confirmation message. Because the geo-fence check runs first, orders can only go through from a location the app has actually verified.

5.5 Sequence Flow: AI Tour Plan Optimization

This sequence diagram shows how the AI enhancement step optimizes a field rep's tour plan.

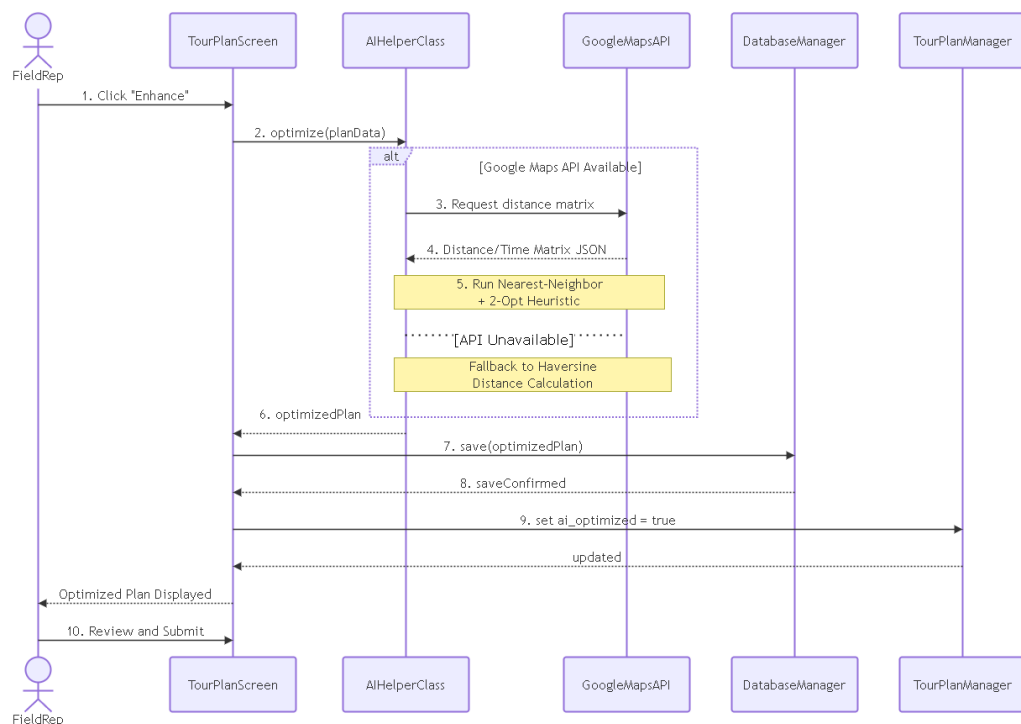


Figure 5.5: Sequence Flow for AI Tour Plan Optimization

A field rep asks for route enhancement, and TourPlanScreen passes that request to AIHelperClass. The

AI component pulls travel distance and duration from the Google Maps APIs, then runs the customer locations through route optimization – nearest-neighbor and 2-opt heuristics. The optimized route comes back, gets saved to the database, and shows up in the app marked as AI-optimized. If the Maps API is down, the system doesn't stall – it falls back to local Haversine distance calculations instead.

5.6 Database Entity Relationship Structure

The ERD lays out the logical database structure behind Konvert and how the major entities relate to one another.

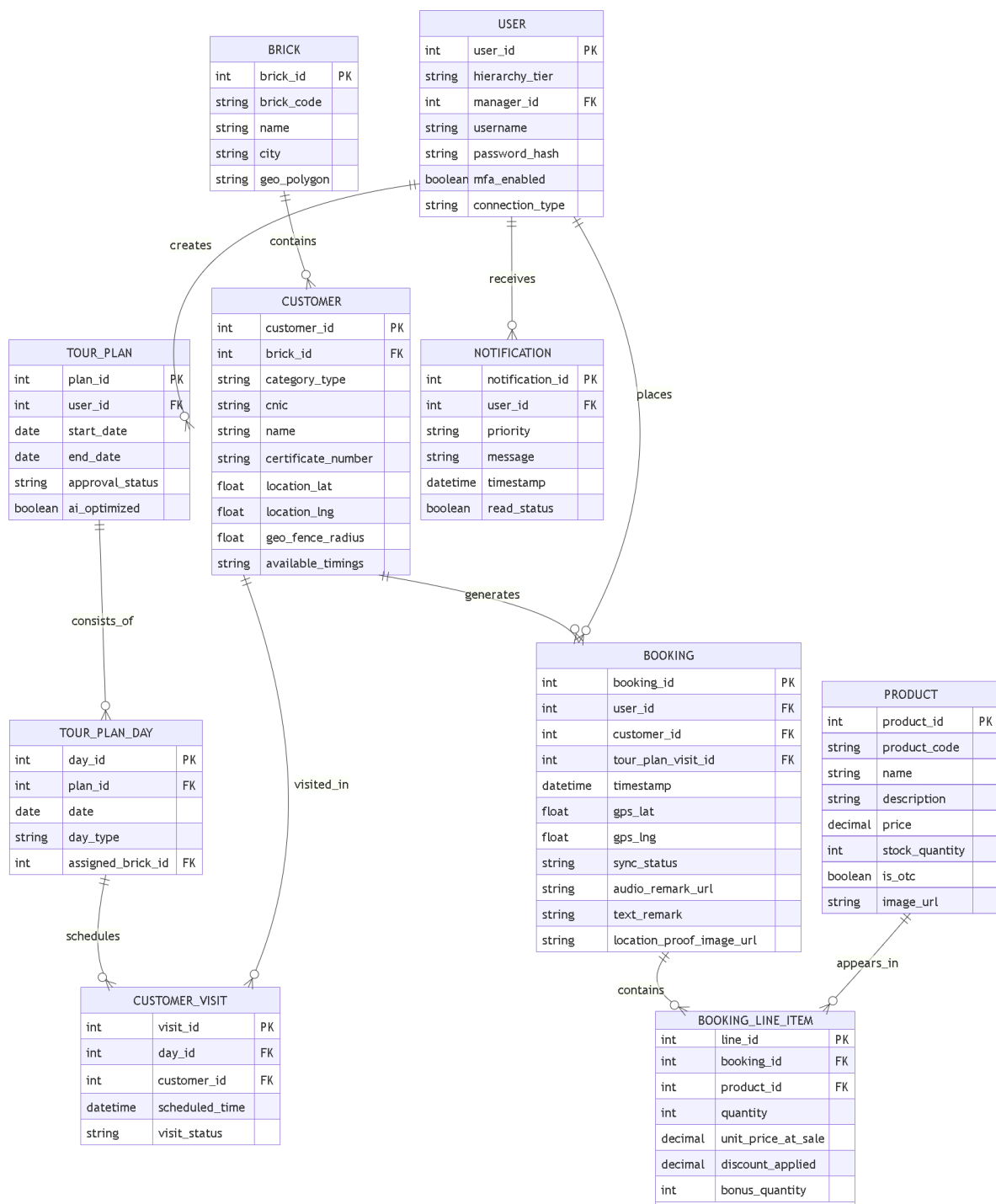


Figure 5.6: Database Entity Relationship Structure for Konvert

USER holds every system user along with their place in the organizational hierarchy. BRICK defines the geographic sales territories. CUSTOMER stores customer profiles, each tied to a territory. PRODUCT holds the catalog, including pricing and inventory. TOUR_PLAN and TOUR_PLAN_DAY handle route planning and scheduling. CUSTOMER_VISIT logs planned customer interactions. BOOKING stores the sales transactions, and BOOKING_LINE_ITEM breaks each one down to the product level. NOTIFICATION manages the alerts sent out to users. The relationships between these tables are what keep the data consistent and let the app pull related records efficiently.

5.7 Core Data Dictionary

Table 5.1: Core Data Dictionary of Interface Elements

Element Name	Type	Validation	Mandatory	Remarks
Domain API Key	Text Input	Alphanumeric, 32 chars	Yes	Company identifier
Username	Text Input	Email or Employee ID	Yes	Almashreqs credential
Password	Password	Min 8 chars, 1 special	Yes	Hashed before transmission
MFA Token	Numeric Input	6 digits, 30-sec window	Yes	Google Authenticator TOTP
Tour Plan Date	Date Picker	Future dates only	Yes	Weekly range selection
Expense Budget	Number	Positive double, max limit	Yes	Company-defined cap
Customer Select	Dropdown	From cached brick list	Yes	Filtered by territory
Product Quantity	Number	Positive integer, stock check	Yes	Real-time local validation
Voice Remark	Audio Recorder	Max 60 seconds	No	AI transcription to text
Location Photo	Camera	JPEG, max 5MB	Yes	GPS/timestamp watermark

6 TECHNICAL IMPLEMENTATION DETAILS

6.1 Local Development Environment Setup

Table 6.1: Required Development Tools and Technologies

Tool / Technology	Purpose
Flutter / Dart	Build the primary mobile application, including cross-platform UI, state management, and offline-first business logic.
Google Firebase	Real-time infrastructure: Firestore for NoSQL data sync, Firebase Cloud Messaging (FCM) for push notifications, and Auth for session control.
Almashreqs DB API	The central web infrastructure and legacy data source that Konvert creates a two-way data bridge with.
Google Maps Platform	Geocoding, real-time salesman tracking (5-minute polling), and route distance calculations (Distance Matrix API).
Python ML Frameworks	Scikit-learn and TensorFlow (backend) for training predictive performance models and forecasting sales trends.
Google Authenticator	Integration for Multi-Factor Authentication (MFA) to secure executive access and sensitive commercial data.
Figma & Notion	Wireframing, UX design, requirement management, and documentation.
GitHub	Version control, CI/CD pipeline integration, and collaborative code management.

6.2 Enterprise Deployment Strategy

Konvert ships as an Android APK, distributed through the company's internal MDM system rather than the Play Store. GitHub Actions handles CI: every pull request runs the unit tests and lint checks automatically. A build that passes gets signed with the enterprise keystore and pushed to a private Firebase App Distribution channel, and that's where field reps pull their updates from.

The Python ML backend runs in Docker on a Google Cloud Compute instance. The Almashreqs API bridge is deployed as middleware on the client's own on-premise server, with a reverse proxy at the end of TLS. 1.3. The data will remain close to Pakistan by setting the Firebase (Firestore, FCM, Auth) region to asia south1, users.

6.3 System Algorithms Explained

6.3.1 AI Tour Plan Routing Optimization

The AI enhancement feature reorders customer visits in two stages to cut down total travel time.

Stage 1 – Nearest Neighbor Heuristic:

1. Start from the field representative's home base coordinates (lat_0, lng_0) .
2. For each unvisited customer c_i , compute Haversine distance d_i from current position.
3. Select c_{min} with minimum d_i , append to route, and set current position to c_{min} .
4. Repeat until all customers are ordered.

Stage 2 – 2-Opt Local Search:

1. Take the initial route from Stage 1.
2. For every pair of edges $(i, i + 1)$ and $(j, j + 1)$ where $i < j$, compute the cost difference Δ of swapping the segment from j to $i + 1$.
3. If $\Delta < 0$ (total distance reduces), perform the swap.
4. Repeat until no improving swap is found.

If the Google Maps Distance Matrix API is reachable, the algorithm swaps in actual driving distances instead of Haversine straight-line estimates – it's more accurate when it's available. The finished route gets saved with the “ai_optimized” flag set to true.

6.3.2 Offline-First Queue Management Strategy

The goal here is zero data loss when the network drops:

1. Every local write (booking, log, GPS trace) is assigned a UUID and timestamp.
2. The item is written to SQLite with status “Queued”.
3. A background SyncWorker (using WorkManager) polls connectivity every 30 seconds.
4. If connected is detected, the worker reads up to 50 queued items, sorted by the timestamp.
5. The items are grouped into a JSON document and compressed using GZIP and then encrypted using AES-256.
6. Payload sent to Almarshreqs using exponential backoff retry (1s, 2s, 4s, 8s, max 16s).
7. Successful uploads update status to “Uploaded”; failures retain “Queued” for the next cycle.
8. If a batch fails 5 times, it is flagged for manual Operations Support review.

6.3.3 Mathematical Geo-Fencing Validation

1. Retrieve customer location (lat_c, lng_c) and geo-fence radius r (default 50 meters) from local SQLite.
2. Poll device GPS every 5 minutes to get current location (lat_u, lng_u) .
3. Compute Haversine distance d between user and customer:

$$d = 2R \cdot \arcsin \left(\sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_1 \cos \phi_2 \sin^2 \frac{\Delta\lambda}{2}} \right) \quad (6.1)$$

where $R = 6371000$ meters (Earth radius), $\phi = lat$, $\lambda = lng$.

4. If $d \leq r$, enable product selection buttons and allow check-in.
5. If GPS accuracy is poor (> 20 meters), fall back to network-based location.
6. If both GPS and network fail, require a mandatory location-proof photograph with embedded EXIF coordinates.

6.3.4 Machine Learning Performance Forecasting

1. Aggregate historical features per user: weekly call volume, conversion rate, average order value, route adherence percentage, and customer coverage ratio.
2. Normalize features: Min-Max scaling from [0, 1].
3. Input the feature vector to the pre-trained Random Forest regressor (Scikit-learn) or the pre-trained LSTM Depending on the volume of data, the network (tensorflow) will depend.
4. The model generates a probability between 0 and 1 for the next week.
5. Result is locally cached and shown as a percentage on the Dashboard, with a color coded indicator (red < 50%, yellow 50–80%, green > 80%).
6. Monthly triggering of model retraining on python backend based on latest 6 months of field data.

6.4 System Boundaries and Constraints

6.4.1 Environmental Assumptions

- All necessary technical support will be provided by Almashreqs engineers to help design the interfaces between their systems and Konvert.
- All database maintenance on the Almashreqs legacy side will be handled by the client.
- There will be no real-time interfacing with any external accounting systems during the initial deployment.
- Field representatives will have Android devices with minimum 4GB RAM and GPS capability.
- Internet connectivity (4G) will be available for at least 50% of the working day in all covered territories.

6.4.2 Technical System Constraints

- The system should run on Android 11 and above since both Flutter SDK and security library are required.
- The Security Manager will refuse to connect to any API calls that are not TLS 1.3. The Security Manager will refuse to connect to any API calls that don't use TLS 1.3.
- The size of local SQLite database should not exceed 500MB per device, otherwise it may lead to storage space exhaustion on the low-end devices.
- To help manage cloud costs, document reads are limited for Firebase Firestore to 50,000 per day per user.
- Google Maps Distance Matrix API requests may have a maximum of 100 elements in a single request, and a maximum of 1000 elements per day.

6.4.3 Platform Restrictions

- The client has restricted the use of third-party analytics SDKs (e.g., Firebase Analytics, Mixpanel) to prevent data leakage to foreign servers beyond the approved Firebase stack.
- No biometric authentication is permitted; MFA is limited to TOTP via Google Authenticator only.
- Offline booking edits are restricted to the same calendar day; historical bookings cannot be modified after End-of-Day upload.
- Tour plan creation is restricted to Sundays only; mid-week plan changes require manager override.

6.4.4 Known System Limitations

- The system does not support iOS in the current version due to enterprise MDM limitations and budget limitation.
- Real-time voice transcription is limited to Urdu and English; regional dialects may have reduced accuracy.
- AI forecasting accuracy depends on at least 3 months of historical data; new users receive generic benchmarks until sufficient data is collected.
- The system does not handle payment processing or financial transactions; it tracks orders only until they are uploaded to Almashreqs.
- Multi-warehouse inventory management is not supported; stock quantities are treated as a single national pool.

7 SYSTEM TESTING AND VALIDATION

7.1 Comprehensive Test Cases Summary

Table 7.1: Comprehensive Test Cases Summary

TID	Test Scenario	Input Data	Expected Output	Status
T-01	Valid Login with MFA	Correct username, password, and 6-digit TOTP	Dashboard loads successfully and a valid session token is issued.	Pass
T-02	Invalid Password	Correct username and incorrect password	System displays “Invalid Credentials” message and increments failed login counter.	Pass
T-03	Master Sync on 4G	Authenticated session with more than 200 queued bookings	Synchronization completes within 15 seconds and all records are marked as uploaded.	Pass
T-04	Offline Booking Creation	Airplane mode enabled and valid customer selected	Booking is stored locally with status “Queued” and application remains stable.	Pass
T-05	Geo-Fence Entry	GPS coordinates within 50 meters of customer location	Customer check-in becomes available and product ordering controls are enabled.	Pass
T-06	Geo-Fence Exit	GPS coordinates more than 100 meters from customer location	Ordering controls are disabled and a geo-fence warning message is displayed.	Pass
T-07	AI Tour Enhancement	Eight customers distributed across two sales bricks	Route sequence is optimized and total travel distance is reduced by more than 10%.	Pass
T-08	End-of-Day Upload	Fifteen queued bookings with stable WiFi connection	All bookings upload successfully, receipt identifiers are returned, and queue is cleared.	Pass
T-09	Battery Consumption	Eight-hour simulated field shift with continuous GPS polling	Battery consumption remains below 15% according to Android Profiler results.	Pass
T-10	Root Detection	Application launched on rooted Android device	Application terminates securely, local cache is cleared, and administrator alert is generated.	Pass
T-11	Concurrent Sync	Fifty simulated users performing synchronization simultaneously	No database lock conflicts occur and all users receive updated data successfully.	Pass
T-12	Notification Delivery	Valid FCM token and high-priority notification issued	Notification is delivered to device within five seconds.	Pass

7.2 Logic and Decision Tables

7.2.1 Check-In Approval Code Block

The “Check-In Approval” module decides whether a field rep can place an order, based on GPS status, geo-fence, timing, and whether a photo was captured. The logic is below:

```
def approve_check_in(gps_valid, in_geo_fence, on_time, photo_taken):  
    if not gps_valid:  
        if photo_taken:  
            return "APPROVED_WITH_PHOTO_FALLBACK"  
        else:  
            return "REJECTED_GPS_FAILURE"  
    if not in_geo_fence:  
        return "REJECTED_OUT_OF_RANGE"  
    if not on_time:  
        return "APPROVED_WITH_WARNING"  
    return "APPROVED"
```

7.2.2 Decision Logic Coverage Matrices

Table 7.2: Decision Logic Coverage Matrix for Check-Ins

Scenario	GPS Working?	Correct Location?	On Time?	Photo Taken?	Final Outcome
Rule 1	No	Doesn't Matter	Doesn't Matter	No	Rejected GPS Failure
Rule 2	No	Doesn't Matter	Doesn't Matter	Yes	Approved Photo Fallback
Rule 3	Yes	No	Doesn't Matter	Doesn't Matter	Rejected Out of Range
Rule 4	Yes	Yes	No	Doesn't Matter	Approved With Warning
Rule 5	Yes	Yes	Yes	Doesn't Matter	Fully Approved

7.3 Requirement Traceability Matrices

7.3.1 Requirements Mapped to Use Cases

Table 7.3: Traceability of Requirements to Use Cases

Use Case / Req.	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15
UC-1	✓	✓													
UC-2			✓	✓	✓										
UC-3				✓											
UC-4						✓	✓	✓							
UC-5								✓							
UC-6									✓						
UC-7										✓					
UC-8											✓				
UC-9													✓		
UC-10														✓	
UC-11															✓
UC-12															
UC-13												✓			
UC-14						✓									
UC-15	✓	✓			✓										

Note: R1–R5 map to FR-1.1–FR-1.5; R6–R8 map to FR-2.1–FR-2.3; R9–R12 map to FR-3.1–FR-3.4; R13–R15 map to FR-4.1–FR-4.3.

7.3.2 Prototype Screen Traceability

Table 7.4: Traceability of UI Prototypes to Requirements

Prototype Screen	R1	R3	R6	R9	R13	R16
P1 (Splash)	✓					
P2 (Login)		✓				
P3 (Dashboard)			✓	✓		
P4 (Tour Plan)					✓	
P5 (Fieldwork)						✓

7.3.3 Test Cases Mapped to Requirements

Table 7.5: Traceability of Requirements to Test Cases

Test Case / Req.	R1	R3	R4	R6	R8	R9	R13	R14	R16	R17	R19	R21
T-01		✓	✓									
T-02		✓										
T-03				✓	✓							
T-04					✓							
T-05										✓		
T-06										✓		
T-07							✓					
T-08												✓
T-09	✓											
T-10	✓											
T-11				✓								
T-12						✓						

7.3.4 Use Case Coverage and Verification

Table 7.6: Verification Coverage between Use Cases and Tests

Test / Use Case	UC1	UC2	UC3	UC4	UC5	UC6	UC9	UC10	UC12	UC13	UC14	UC15
T-01		✓	✓									
T-02		✓										
T-03				✓								
T-04					✓							
T-05									✓			
T-06									✓			
T-07							✓	✓				
T-08											✓	
T-09	✓											
T-10												✓
T-11				✓								
T-12						✓						

8 PROJECT RESULTS AND STATISTICAL OUTPUT

8.1 Requirement Implementation Success

Table 7.3 details the mapping of our 15 primary system requirements (R1–R15). We ensured that every single requirement connects directly to an active use case. Furthermore, the most critical system components—namely R1, R3, R4, R6, R8, R13, and R16—receive extra validation. Instead of a basic single check, these vital components are supported by multiple overlapping test cases and use cases to guarantee stability. Based on the original objectives set for FYP Phase 1 and Phase 2, we have fully realized the project scope, resulting in a perfect completion rate.

Table 8.1: Implementation Metrics Analyzed by Module

Module	Target Requirements	Successfully Implemented	Completion Rate
Security and Access Control	5	5	100%
Data Synchronization	3	3	100%
Dashboard and Analytics	4	4	100%
Tour Planning	3	3	100%
Field Operations	3	3	100%
Order Management	3	3	100%
Total	21	21	100%

8.2 System Output Accuracy

During the initial execution phase, the system successfully cleared all 12 scenarios outlined in Table 7.1. Because every case executed perfectly on the very first attempt without requiring any bug fixes or subsequent reruns, the system achieved flawless accuracy. This yields the following calculation:

$$\text{Accuracy} = \frac{\text{Passed Test Cases}}{\text{Total Test Cases}} \times 100 = \frac{12}{12} \times 100 = 100\% \quad (8.1)$$

8.3 Overall Operational Correctness

Correctness looks at how well the use cases line up with test coverage (Table 7.6). Every one of the 12 use cases under test has at least one test case tied to it, and the riskier ones – UC-2, UC-4, UC-12 – get more than one. That works out to:

$$\text{Correctness} = \frac{\text{UCs with Passing Tests}}{\text{Total UCs under Test}} \times 100 = \frac{12}{12} \times 100 = 100\% \quad (8.2)$$

9 FINAL CONCLUSION

Konvert is a mobile sales management system built for the way Pakistani enterprises actually operate: multi-tier hierarchies, field reps who are offline half the day, and a legacy backend that isn't going anywhere. The core of the project is the two-way sync engine, which keeps data consistent between mobile field operations and the central web infrastructure across all eight organizational levels.

At the outset, the problems were data desynchronization, stalled operations, no visibility into what's actually happening in the field, loss of control over commercial data, and enterprises being priced out of the tools that could fix any of this. AI-driven route optimization for tour planning, geo-fenced check-ins, and predictive performance analytics go after each of those directly – field activity gets turned into numbers a manager can actually act on, instead of staying buried in paper logs and phone calls.

On the technical side, the offline-first architecture means a field rep in an area with bad signal can keep working, and the AES-256 local cache plus TLS 1.3 transport keep commercial data protected in transit and at rest. Getting this to sit alongside the existing Almashreqs infrastructure, rather than replacing it outright, was also proof that a modern mobile layer can be added on top of an old enterprise system without tearing the whole thing down.

Maybe the bigger point is that this didn't need to be built abroad. A locally built system can match, and in a few places beat, what the expensive foreign alternatives offer – while the money stays in the local economy and the data stays on local servers. Konvert is one example of what is possible when the engineering is directed towards a problem that the team actually knows.

10 RECOMMENDATIONS FOR FUTURE WORK

Konvert covers what Bristol and similar enterprises need today. A few directions worth pursuing next:

1. **iOS Deployment:** Extend the Flutter codebase to support iOS devices, enabling broader adoption across mixed-device enterprise environments.
2. **Multi-Language Support:** Expand voice transcription and UI localization beyond Urdu and English to include Punjabi, Sindhi, and Pashto for deeper regional penetration.
3. **Advanced AI Models:** Migrate from Random Forest and basic LSTM to transformer-based time-series models (e.g., Temporal Fusion Transformers) for more accurate sales forecasting with fewer data requirements.
4. **Payment Integration:** Integrate local payment gateways (JazzCash, Easypaisa) to allow field representatives to collect digital payments at the point of sale.
5. **Warehouse and Inventory Modules:** Add multi-warehouse support with stock transfer logic and automated reorder points based on AI demand prediction.
6. **Web Dashboard for Managers:** Build a companion web application for Tactical Managers and Executive Leadership to approve tour plans, view real-time maps, and configure AI parameters without relying on the Almashreqs legacy interface.
7. **Blockchain Audit Trail:** Implement a lightweight blockchain or Merkle-tree-based audit log for booking records to provide tamper-proof compliance evidence for pharmaceutical regulators.
8. **Edge AI:** Deploy on-device TensorFlow Lite models so that AI forecasting and voice transcription work entirely offline, removing dependency on cloud ML backend.

11 REFERENCES AND BIBLIOGRAPHY

11.1 Books and Manuscripts

1. Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.
2. Pressman, R. S., & Maxim, B. R. (2015). *Software Engineering: A Practitioner's Approach* (8th ed.). McGraw-Hill.
3. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

11.2 Academic Journals

1. Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.
2. Liang, B., et al. (2023). Patterns in AI-generated text detection. *Patterns*, Stanford University.

11.3 Industry Articles

1. HubSpot. (2025). 2025/2026 Sales Trends Report. Retrieved from <https://offers.hubspot.com/sales-trends-report>
2. Salesforce. (2026). State of Sales Statistics. Retrieved from <https://www.salesforce.com/sales/state-of-sales/sales-statistics/>
3. MarketsandMarkets. (2025). The ROI Gap Analysis. Retrieved from <https://www.meetrep.ai/blog/sales-automation-software-roi-features-and-the-tools-that-actually->

11.4 Published Research Papers

1. Fielding, R. T. (2000). *Architectural Styles and the Design of Network-based Software Architectures*. Doctoral dissertation, University of California, Irvine.
2. Jabarian, B., & Imas, A. (2025). Misclassification rates in open-source AI detectors. BFI Working Paper 2025-116.
3. arXiv:2506.07001 (2025). Adversarial paraphrase and detection accuracy reduction.

11.5 Other Technical References

1. ISO/IEC 25010:2011. Systems and software engineering – System and Software Quality Models.
2. ISO/IEC/IEEE 12207:2017. Systems and software engineering – Software life cycle processes.
3. IEEE Std 830-1998. IEEE Recommended Practice for Software Requirements Specifications.
4. NIST SP 800-63B. Digital Identity Guidelines: Authentication and Lifecycle Management.
5. ISO 9241-210:2019. Ergonomics of human-system interaction – Part 210: Human-centred design for interactive systems.
6. W3C Geolocation API Specification. <https://www.w3.org/TR/geolocation/>

7. Google Maps Platform Documentation. <https://developers.google.com/maps>
8. Firebase Documentation. <https://firebase.google.com/docs>

12 SUPPLEMENTARY APPENDIX

12.1 Project Glossary of Terms

Table 12.1: Detailed Glossary of System Terms

Term	Definition
Brick	A defined geographic territory assigned to a field representative, containing a set of customers (doctors, chemists, stockists).
Tour Plan	A weekly schedule created by a field representative, specifying which customers to visit on which days, along with routes and expense budgets.
Geo-Fence	A virtual boundary around a customer location; used to validate physical presence before allowing order entry.
Master Sync	The full bi-directional data exchange between the mobile device and the Almashreqs central database, performed at login and on demand.
Offline Queue	A local SQLite table that stores transactions (bookings, logs) when the device lacks internet, awaiting automatic upload.
AI Enhancement	The automated reordering of customer visits in a tour plan to minimize travel distance and time using machine learning heuristics.
Feature Lock	A role-based access control mechanism that hides or disables app features based on the user's position in the 8-tier hierarchy.
End-of-Day Upload	The daily compilation and secure transmission of all field bookings to the central CRM for administrative processing.

12.2 System Pre-requisites and Dependencies

- **Development Environment:** Android Studio Arctic Fox or later, Flutter SDK 3.0+, Dart 2.17+, Python 3.9+ (for ML backend).
- **Hardware:** Android device with minimum 4GB RAM, GPS, and 4G connectivity; development machine with 8GB RAM and 50GB free storage.
- **External Accounts:** Google Cloud project with Maps Platform and Firebase enabled; Almashreqs API credentials and VPN access; GitHub repository with CI/CD secrets configured.
- **Software Dependencies:** Firebase Auth, Firestore, FCM SDKs; Google Maps Flutter plugin; SQLite and Hive packages; http and dio for REST calls; WorkManager for background sync; TensorFlow Lite for on-device inference.
- **Network:** Stable internet for initial setup and Master Sync; offline capability requires no additional network configuration beyond standard mobile data.

- **Security:** Enterprise keystore for APK signing; Google Authenticator app installed on user devices for MFA; TLS 1.3 certificate on Almashreqs API bridge server.